Alexandros Angelakis

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Experience

Speech Signal Processing Laboratory, UoC, CSD Heraklion, Greece
Graduate Research Assistant June 2024 - Nov. 2026

- Architected and trained robust diagnostic models to detect Tuberculosis in cough audio, leveraging TensorFlow and Scikit-learn for signal processing and classification.
- Contributed to the development and evaluation of a novel, high-quality, multi-speaker Greek speech dataset; trained and optimized state-of-the-art TTS architectures, including FastSpeech2, VITS, Tacotron2 and HiFi-GAN, using the ESPnet framework. Dataset: huggingface.co/datasets/aangelakis/STOMA

IACM, Foundation of Research and Technology, Hellas Heraklion, Greece
Graduate Research Assistant July 2022 - Nov. 2026

- Lead developer of NeuroDiMe, a Python library for neural-based statistical distance estimation (e.g., f-divergences, IPMs) within a unified framework supporting PyTorch, JAX, and TensorFlow. Deployed in cutting-edge AI workflows including β-VAE representation learning and GAN/VAE generative modeling. Supported by Apple Inc. through IACM since June 2024 (ending June 2025).
- Designed and implemented graph traversal algorithms for optimal sensor placement in Water Distribution Networks, enabling accurate leak localization (Python, GeoPandas, NetworkX, GIS).

Education

University of Crete Heraklion, Greece
MSc. Computer Science and Engineering, Grade: 9.64/10.00 (Excellent) Feb. 2024 - Nov. 2026

- Thesis: "Adaptive AM-FM Features for Parkinson's Disease Detection from Speech", supervised by Dr. George P. Kafentzis
- Specialization: Artificial Intelligence, Machine Learning, Signal Processing.
- Teaching Assistant: Applied Mathematics for Engineers, Digital Signal Processing, Digital Speech Signal Processing.

University of Crete Heraklion, Greece
BSc. Computer Science, Grade: 8.77/10.00 (Excellent) Sept. 2019 - July 2023

- 4x IKY National Excellence Scholarship recipient (2020-2023)
- Chrysanthos and Anastasia Karidis Scholarship for top performance in national entrance exams. (2021)
- Teaching Assistant: Applied Mathematics for Engineers, Probabilities, and Information System Analysis and Design.

Publications

1. **Alexandros Angelakis**, George P. Kafentzis. Adaptive AM-FM Decomposition of Speech for Parkinson's Disease Classification, IberSPEECH 2026.

2. Michail Raptakis, Stavroula Maria Moutzouri, Christos Giannaros, Anastasia Kechagia, Olympia Simantiraki, **Alexandros Angelakis**, Yannis Stylianou, George P. Kafentzis, Yannis Pantazis. STOMA: A Multi-Speaker Greek Speech Corpus, *submitted for publication, 2026 (under review).*

Skills

Programming: Python, MATLAB, C, Java, Bash
Frameworks: PyTorch, Tensorflow, JAX, Numpy, Pandas, Scikit-Learn, Scipy
Tools & Databases: Git, Linux, SQL, PostgreSQL
Domain Expertise: Digital Speech Processing, Generative Modeling, Text-to-Speech (TTS)
Languages: English (Proficient), Greek (Native)

Selected Projects

Diabetes Prediction with Machine Learning. Built an end-to-end AutoML pipeline for real-world diabetes prediction, integrating class balancing, model selection, and nested cross-validation for robust performance (Python, Scikit-Learn, 2025).

Digital Speech Processing Projects. Completed for the Digital Speech Signal Processing course taught by Prof. Yannis Stylianou. MATLAB-based implementations covering voice activity detection, LPC coding, sinusoidal modeling, and speaker identification. Projects involve hands-on speech signal analysis, enhancement, and classification using real-world audio datasets (2023).

AlphaCompiler. A complete compiler and virtual machine for the Alpha programming language (JavaScript-like syntax), developed from scratch (C, Lex, Yacc/Bison, 2022).